inrCM.py — Changelog

Cross-Modal INR Steganography · IUST Kashmir · Group 02

B.Tech Final Year Project · Sem 8 · April 2026

4 Bug Fixes	2 Improvements	3 New Features	9 Total Changes

01 Audio Normalization Bug

BUG FIX

The reconstructed audio had completely wrong amplitude, making it sound distorted or silent. The root cause was that **orig_min** and **orig_max** were captured *after* normalizing the waveform, so they always held values near 0.0 and 1.0 — making the denormalization in **save_audio_modality** a complete no-op.

Before

```
y = (y - y.min()) / (y.max() - y.min() + 1e-8)
meta = {..., 'orig_min': float(y.min()), ...} # y already [0,1]!
```

After

```
orig_min = float(y.min()) # captured BEFORE normalize
orig_max = float(y.max())
y = (y - orig_min) / (orig_max - orig_min + 1e-8)
```

02 Text Encoding / Decoding

BUG FIX

Extracted text came out as garbled random characters. Three compounding issues: (1) characters were encoded as **ord(c) / 127.0** but decoded with clamp to [32, 126], destroying newlines, tabs and many punctuation chars; (2) the decode range was too narrow; (3) **max_chars** was only 1024, silently truncating long files like the Bee Movie script. Replaced with a clean symmetric encode/decode pair spanning ordinals 9–126.

- Encode: (ord(c) - 9) / 117.0 maps range [tab..~] to [0, 1]
- Decode: round(v * 117) + 9 recovers exact ordinal
- max_chars raised from 1024 to 4096

03 Output Head Architecture

IMPROVEMENT

Both output heads used **nn.Sequential(Linear, Sigmoid())** with the activation baked inside the module. Refactored to plain **nn.Linear** heads with **torch.sigmoid()** applied explicitly in the forward methods. Functionally equivalent but cleaner, and allows per-modality activation overrides in future iterations.

04 Added m4a Audio Support

FEATURE

The **detect_modality()** function did not recognise **.m4a** files, forcing manual conversion to MP3 before every run. Added **.m4a** to the audio extensions list.

Before

```
if ext in [".wav", ".mp3", ".flac", ".ogg", ".aac"]:
```

After

```
if ext in [".wav", ".mp3", ".flac", ".ogg", ".aac", ".m4a"]:
```

05 Output Files Overwriting Each Other

BUG FIX

Every run wrote to the same flat **output/** directory, so running audio-in-image followed by text-in-image would silently clobber the first run's files including the weights. Both **hide** and **extract** modes now create a unique timestamped subfolder per run.

- hide mode: output/cover_image__secret_text__20250428_143022/
- extract mode: output/extract_cover_image__secret_audio__20250428_150001/

06 Adaptive Loss Weights

BUG FIX

The loss was hardcoded at **0.6 cover / 0.4 secret** regardless of modality. Text requires near-perfect reconstruction (a single wrong float maps to the wrong character), so giving the secret only 40% weight caused systematic corruption. Weights are now derived from per-modality base scores and normalised to sum to 1.0.

Modality	Base Weight	As Secret (example: img cover)
`image / video`	`1.0`	`50%`
`audio`	`1.5`	`60%`
`text`	`2.5`	`71%`

07 Quality Presets Upgraded

IMPROVEMENT

The old **low** preset (1000 steps, hidden dim 128) produced only ~34k parameters — far too few to memorise 4096 text characters or 22050 audio samples. All presets have been significantly upgraded.

Preset	Steps (old→new)	Hidden Dim (old→new)
`low`	`1,000 → 3,000`	`128 → 256`
`medium`	`2,000 → 6,000`	`256 → 512`
`high`	`5,000 → 12,000`	`512 → 512`
`ultra`	`10,000 → 20,000`	`1024 → 1024`

08 PSNR Verdict in report.json

FEATURE

The report now includes **verdict_cover** and **verdict_secret** fields with human-readable quality assessments based on per-modality PSNR thresholds. Also added **loss_weight_cover** and **loss_weight_secret** for full reproducibility.

- text: GOOD >= 38 dB | OK >= 30 dB | POOR >= 20 dB | UNUSABLE < 20
- audio: GOOD >= 35 dB | OK >= 25 dB | POOR >= 15 dB | UNUSABLE < 15
- image/video: GOOD >= 35 dB | OK >= 28 dB | POOR >= 18 dB | UNUSABLE < 18

09 Live PSNR Warning After Training

FEATURE

Previously, a low PSNR secret could only be discovered by inspecting the output files after the fact. The script now immediately prints a **warning** to the console right after training completes if the secret PSNR falls below the minimum viable threshold for the modality, and advises re-running at a higher quality preset.

- Threshold: text >= 30 dB, audio >= 25 dB, others >= 28 dB
- Prints: ■ Secret PSNR too low (10.5 dB < 30 dB threshold).
- Prints: Output will be corrupted. Re-run with --quality medium or higher.